



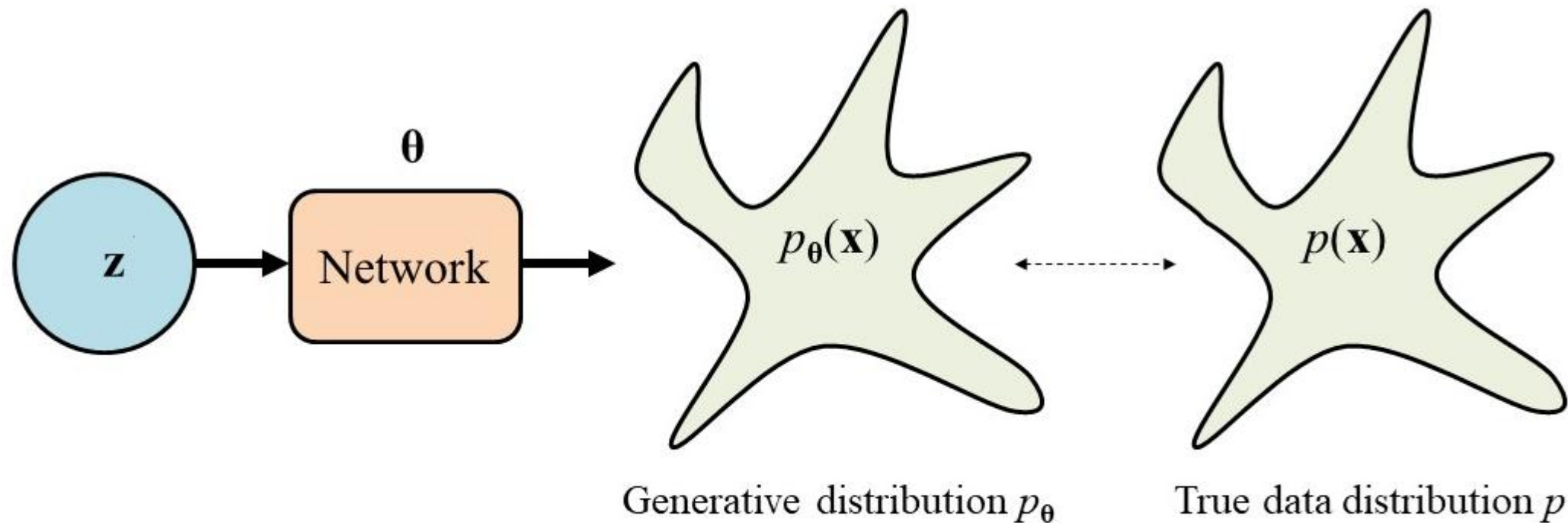
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Deep Learning and Applications

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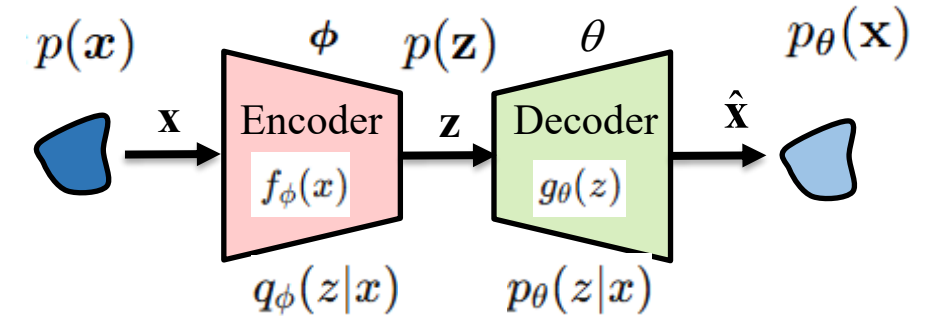
Generative model

- Given observed samples x from a distribution of interest, the goal of a generative model is to learn to model its true **data distribution $p(x)$** .



Maximum log likelihood $\rightarrow \log p_{\theta}(x)$

Variational Autoencoder (VAE)



- Maximum likelihood objective

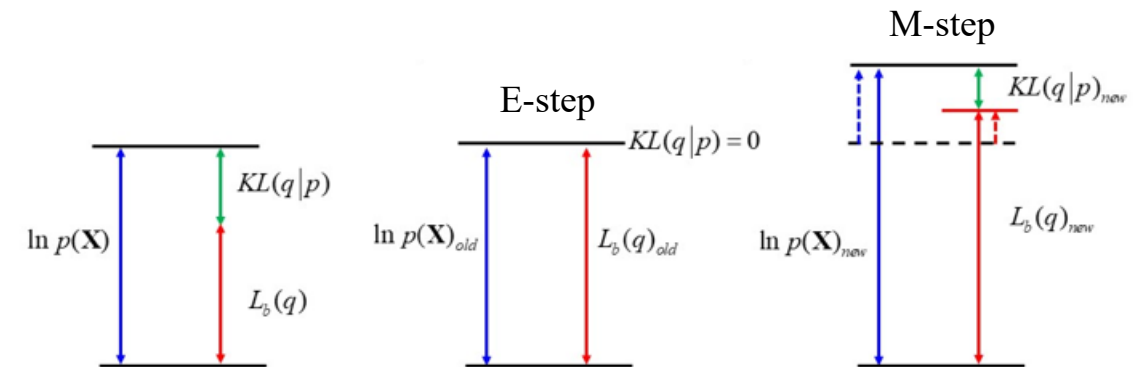
$$\log p_{\theta}(x) = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - D_{KL}(q_{\phi}(z|x)||p(z)) + D_{KL}(q_{\phi}(z|x)||p_{\theta}(z|x))$$

The decoder learns to reconstruct the data Prevents the encoder from mapping each data point to its own isolated region in latent space. The encoder is learning to find the right z that makes sense of x .

- Training Dynamics (Likelihood Trend)
 - ELBO gradually increases as encoder/decoder start coordinating.
 - KL divergence shrinks — encoder approaches true posterior.
 - ELBO continues improving; log-likelihood typically rises as the model better fits the data.

EM algorithm

an iterative optimization method that alternates between an Expectation step (E-step) and a Maximization step (M-step) to continuously increase the Evidence Lower Bound (ELBO), thereby improving the log-likelihood of the data.



→ EM is an iterative method to increase the lower bound which consequently increases the data likelihood

Log-likelihood may not increase every epoch (not monotonic).
The overall trend of increasing ELBO → higher likelihood over time.

Backward pass

- Update encoder parameters ϕ (fix θ)
- Update decoder parameters θ (fix ϕ)



Generative Model Design Paradigms

- **Variational Autoencoders (VAE)**

1. Utilize *variational probabilistic inference* with latent variables
2. Support explicit **likelihood modeling**
3. Often produce **blurry** or less detailed samples

- **Generative Adversarial Networks (GAN)**

1. Rely on **adversarial training**
2. Capable of highly realistic, sharp sample generation
3. Can suffer from training instability and **mode collapse** (failing to capture the full diversity)

$$\text{Support}(p_G) \subset \text{Support}(p_{data})$$

- **Diffusion Models**

Combine:

Likelihood-based, probabilistic modeling (**VAE-like**)

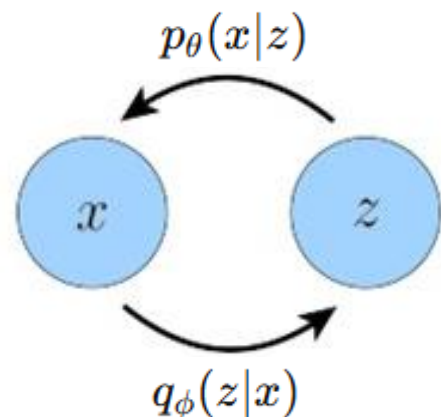
High-fidelity sample generation (**GAN-like**)

Now among the most prominent generative modeling frameworks

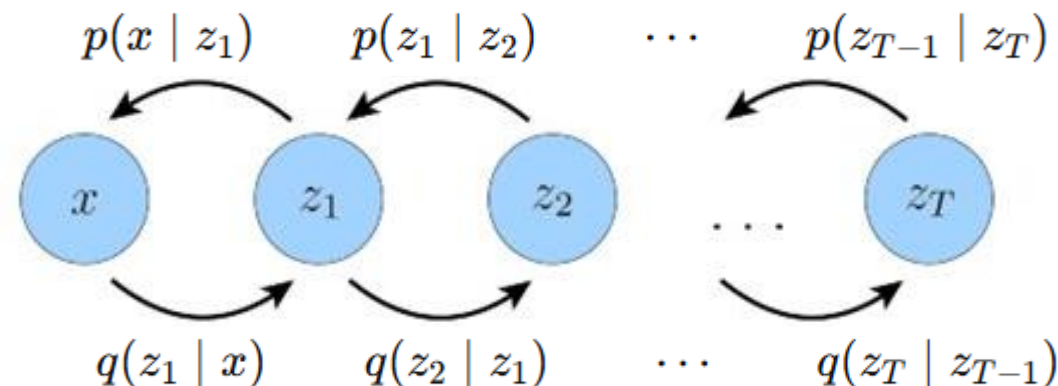


Diffusion Models

VAE

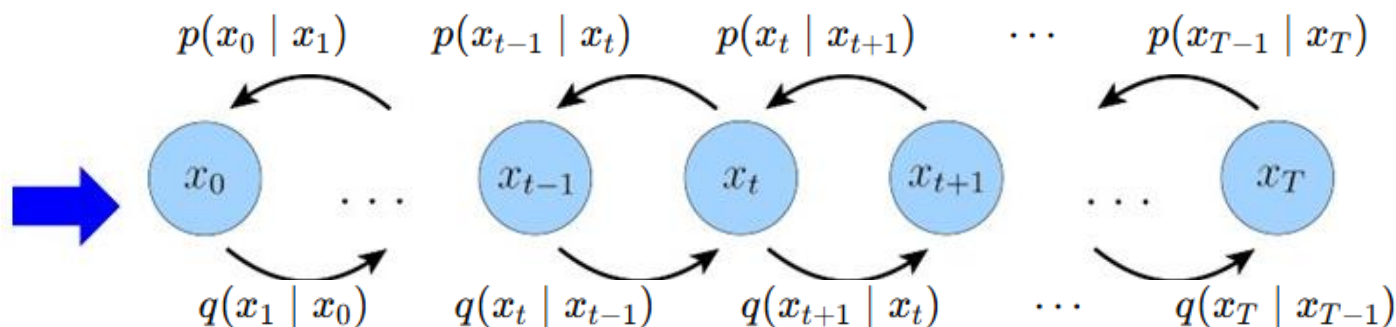


Hierarchical VAE



Decoding each latent z_t only conditions on previous latent z_{t+1}

Diffusion Model/ Variational Diffusion Model (VDM)



Variational diffusion gradually destroys structure in data:

$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_T \sim \mathcal{N}(0, I)$$



x_0
Clean data



x_t
 $\rightarrow \rightarrow \rightarrow \rightarrow$



x_T
Noisy data

- The latent dimension equals the data dimension
- The latent encoder is not learned — fixed noise schedule
- The final latent distribution is standard Gaussian $p(x_T) = \mathcal{N}(0, I)$

Diffusion Models

- Training objective is derived from a **variational ELBO**, just like VAEs

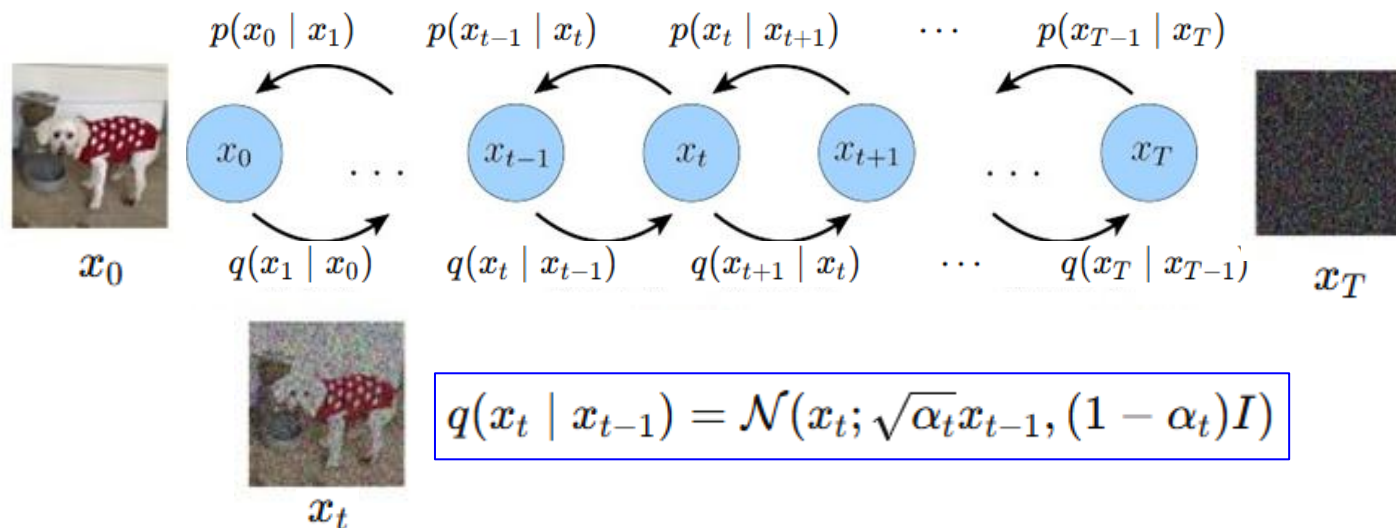
(Variational Diffusion Model, VDM)

1. Forward noising process $q(x_1, x_2, \dots, x_T | x_0)$ acts as a fixed variational distribution

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}) \leftrightarrow \text{VAE Encoder } q_\phi(z | x)$$

2. Reverse denoising process $p_\theta(x_0, x_1, \dots, x_{T-1} | x_T)$ is the model we optimize.

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t) \leftrightarrow \text{VAE Decoder } p_\theta(x | z)$$



- Noise Retention Factor α_t

$$x_t \approx \sqrt{\alpha_t} x_{t-1} + \text{noise}$$

Markov chain: Only the present affects the future, not the past.
- Each step **fades the signal** by $\sqrt{\alpha_t}$
- Each step **adds Gaussian noise** with variance $1 - \alpha_t$
- This predefined process is **not learned**.
 (reverse process is the only part that must be learned.)



Diffusion Models

- Training objective is derived from a variational ELBO, just like VAEs
- Goal: Maximum likelihood

$$p_{\theta}(x_0) = \int p_{\theta}(x_{0:T}) dx_{1:T} = \int p(x_T) \prod_{t=1}^T p_{\theta}(x_{t-1} | x_t) dx_{1:T} \quad \text{intractable}$$

Introduce a fixed variational distribution $q(x_{1:T} | x_0)$

$$\log p_{\theta}(x_0) = \log \int q(x_{1:T}|x_0) \frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} dx_{1:T}$$

Apply the standard variational trick

$$p_{\theta}(x_0) = \int p_{\theta}(x_{0:T}) dx_{1:T} = \int q(x_{1:T}|x_0) \frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} dx_{1:T}$$

An expectation over q

$$\log p_{\theta}(x_0) = \log \int q(x_{1:T}|x_0) \frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} dx_{1:T} = \log \mathbb{E}_q \left[\frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} \right]$$

Jensen's Inequality

$$\log p_{\theta}(x_0) \geq \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} \right] = \mathbb{E}_q [\log p_{\theta}(x_{0:T}) - \log q(x_{1:T}|x_0)] \quad \text{ELBO}$$

$$\begin{cases} q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}) \\ \text{Each term is predefined} \\ q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I) \end{cases}$$

Jensen's Inequality
for the concave function $\log[\cdot]$,

$$f(\text{average}) \geq \text{average}(f)$$

$$f(\mathbb{E}_q[z]) \geq \mathbb{E}_q[f(z)]$$

$$\log \mathbb{E}_q[z] \geq \mathbb{E}_q[\log z]$$



Diffusion Models

- Training objective is derived from a variational ELBO, just like VAEs
- Goal: Maximum likelihood

$$\log p_{\theta}(x_0) \geq \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} \right] = \mathbb{E}_q [\log p_{\theta}(x_{0:T}) - \log q(x_{1:T}|x_0)]$$

Generative model

$$\log p_{\theta}(x_{0:T}) = \log p(x_T) + \sum_{t=1}^T \log p_{\theta}(x_{t-1} | x_t)$$
$$\sum_{t=1}^T \log p_{\theta}(x_{t-1} | x_t) = \log p_{\theta}(x_0 | x_1) + \sum_{t=2}^T \log p_{\theta}(x_{t-1} | x_t)$$

$$\log p_{\theta}(x_{0:T}) = \log p(x_T) + \log p_{\theta}(x_0 | x_1) + \sum_{t=2}^T \log p_{\theta}(x_{t-1} | x_t)$$

Forward diffusion

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1})$$

$$\log q(x_{1:T} | x_0) = \log q(x_1 | x_0) + \sum_{t=2}^T \log q(x_t | x_{t-1}) = \log q(x_{1:T} | x_0) = \log q(x_T | x_0) + \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)$$



Diffusion Models

- Training objective is derived from a variational ELBO, just like VAEs
- Goal: Maximum likelihood

$$\log p_{\theta}(x_{0:T}) = \log p(x_T) + \log p_{\theta}(x_0 | x_1) + \sum_{t=2}^T \log p_{\theta}(x_{t-1} | x_t)$$

$$\log q(x_{1:T} | x_0) = \log q(x_T | x_0) + \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)$$

$$\mathcal{L} = \mathbb{E}_q [\log p_{\theta}(x_{0:T}) - \log q(x_{1:T} | x_0)]$$

$$= \underbrace{\mathbb{E}_q [\log p_{\theta}(x_0 | x_1)]}_{\text{reconstruction term}} + \underbrace{\mathbb{E}_q [\log p(x_T) - \log q(x_T | x_0)]}_{\text{prior matching term}} + \sum_{t=2}^T \log p_{\theta}(x_{t-1} | x_t) - \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)$$

Denoising matching terms (for t = 2...T)

VAE

$$\mathcal{L}_{\text{ELBO}}(\theta, \phi) = \underbrace{\mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x | z)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_{\phi}(z | x) \| p(z))}_{\text{prior matching term}}$$

Diffusion Models

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_q [\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)] \\ &= \underbrace{\mathbb{E}_q [\log p_\theta(x_0 | x_1)]}_{\text{reconstruction term}} + \underbrace{\mathbb{E}_q [\log p(x_T) - \log q(x_T | x_0)]}_{\text{prior matching term}} + \sum_{t=2}^T \log p_\theta(x_{t-1} | x_t) - \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)\end{aligned}$$

Denoising matching terms (for t = 2...T)

- Reconstruction term
 1. In reverse diffusion, the model must recover the clean image x_0 from a noisy observation x_1 .
 2. Reverse process is probabilistic, not deterministic.

$$p_\theta(x_0 | x_1)$$

3. The model predicts the **mean** of this distribution: $\hat{x} = \mathbb{E}_{p_\theta}[x_0 | x_1]$
4. Maximizing the likelihood of this Gaussian is equivalent to minimizing mean square error.

L2 loss (MSE) $\|x_0 - \hat{x}\|_2^2$

- Prior matching term (No trainable parameters)

$$\mathbb{E}_q [\log p(x_T) - \log q(x_T | x_0)] = -D_{\text{KL}}(q(x_T | x_0) \| p(x_T)) \quad \text{model's prior} \quad p(x_T) = \mathcal{N}(0, I) \quad \nabla_\theta D_{\text{KL}} = 0$$

match the prior used by the generative model.



Diffusion Models

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_q [\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)] \\ &= \underbrace{\mathbb{E}_q [\log p_\theta(x_0 | x_1)]}_{\text{reconstruction term}} + \underbrace{\mathbb{E}_q [\log p(x_T) - \log q(x_T | x_0)]}_{\text{prior matching term}} + \underbrace{\sum_{t=2}^T \log p_\theta(x_{t-1} | x_t) - \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)}_{\text{Denoising matching terms (for } t = 2 \dots T)}\end{aligned}$$

- Denoising matching terms (for $t = 2 \dots T$)

$$\sum_{t=2}^T \mathbb{E}_q [\log p_\theta(x_{t-1} | x_t) - \log q(x_t | x_{t-1}, x_0)]$$

Model's reverse step Forward diffusion's true posterior

Extracting the t -th Term

$$\mathbb{E}_q [\log p_\theta(x_{t-1} | x_t) - \log q(x_{t-1} | x_t, x_0)] = -D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t))$$

This KL forces the model's reverse transition to imitate the forward diffusion's true posterior

$$\sum_{t=2}^T D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t))$$

At $t = 1$?



1. The model learns how to slightly denoise at each timestep.
2. This learning objective naturally becomes minimizing a Gaussian KL.

Diffusion Models

- Denoising matching terms

$$\mathcal{L}_{\text{denoise}} = - \sum_{t=2}^T \mathbb{E}_{q(x_t|x_0)} \left[D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_{\theta}(x_{t-1} | x_t)) \right]$$

1. KL between two Gaussians in d dimensions:

$$\begin{aligned} D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_{\theta}(x_{t-1} | x_t)) \\ = \frac{1}{2} \left[\log \frac{\det \Sigma_{\theta}(t)}{\det \Sigma_q(t)} - d + \text{tr}(\Sigma_{\theta}(t)^{-1} \Sigma_q(t)) \right. \\ \left. + (\mu_{\theta}(x_t, t) - \mu_q(x_t, x_0))^{\top} \Sigma_{\theta}(t)^{-1} (\mu_{\theta}(x_t, t) - \mu_q(x_t, x_0)) \right] \end{aligned}$$

2. Fix the covariance (practical choice)

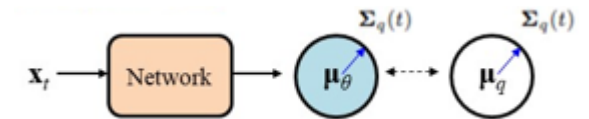
$$\Sigma_{\theta}(t) = \Sigma_q(t) = \sigma_t^2 I \quad (\text{known scalar schedule})$$

$$\log \frac{\det \Sigma_{\theta}}{\det \Sigma_q} = 0$$

$$\text{tr}(\Sigma_{\theta}^{-1} \Sigma_q) = d$$

$$\log \frac{\det \Sigma_{\theta}}{\det \Sigma_q} - d + \text{tr}(\Sigma_{\theta}^{-1} \Sigma_q) = 0.$$

$$\begin{aligned} q(x_{t-1} | x_t, x_0) &= \mathcal{N}(\mu_q(x_t, x_0), \Sigma_q(t)), \\ p_{\theta}(x_{t-1} | x_t) &= \mathcal{N}(\mu_{\theta}(x_t, t), \Sigma_{\theta}(t)). \end{aligned}$$



Hard to Train

- Model must learn **both the mean (μ) and the covariance (Σ)**
- Training is **highly sensitive** to Σ and often **unstable**
- This difficulty occurs at **every single diffusion time step**

$$D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_{\theta}(x_{t-1} | x_t)) = \frac{1}{2\sigma_t^2} \|\mu_{\theta}(x_t, t) - \mu_q(x_t, x_0)\|_2^2.$$

$$\mathcal{L}_{\text{denoise}} = - \sum_{t=2}^T \mathbb{E}_{q(x_t|x_0)} \left[\frac{1}{2\sigma_t^2} \|\mu_{\theta}(x_t, t) - \mu_q(x_t, x_0)\|_2^2 \right]$$

An MSE loss between the model mean μ_{θ} and the true posterior mean μ_q at each timestep.